

Infrastructure and Environmental Assessment

ENGR49580D Technology, Society and Sustainability

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in·fra·struc·ture
/'ɪnfərəˌstræk(t)ʃHər/
noun

the basic physical and organizational structures and facilities (e.g. buildings, roads, power supplies) needed for the operation of a society or enterprise. "the social and economic infrastructure of a country"

Infrastructure

Types of Infrastructure

- Infrastructure may be broadly divided into two categories:
 - Physical Infrastructure
 - Roads, bridges, power lines, sewers
 - Institutional Infrastructure
 - Governments, banks, stock exchanges

This lecture will focus on physical infrastructure

Infrastructure

Who owns it?

- There are a number of ownership models for infrastructure
 - Government owned
 - Government Run
 - Authority / Agency / Commission Run
 - Crown Corporation
 - Privately owned
 - Public-Private Partnership (P3 or PPP)
 - Ownership transfer

Infrastructure

Government Owned Infrastructure - municipal

- 60% of infrastructure in Canada is owned by local governments†
- Typically these local governments are responsible for infrastructure maintenance, but do not always have the means to construct new projects on their own

† <https://fcm.ca/en/focus-areas/infrastructure>

Infrastructure

Government Owned Infrastructure - Municipal

- Infrastructure capital costs are typically funded by:
 - Partnerships with provincial / federal governments
 - Development fees
 - Debt
- Infrastructure operating and maintenance costs may be paid for by
 - Taxes
 - User Fees

Infrastructure

Government Owned Infrastructure - Provincial / Federal

- Larger governments take on larger infrastructure projects such as
 - International Airports
 - Freeways
 - Regional and national rail
- These governments also take on smaller projects in areas where the local population is not large enough to sustain them such as
 - Arterial roads in northern regions

Infrastructure

Government Owned Infrastructure Operators

- Government owned infrastructure may be operated in a number of ways:
 - By government departments / ministries
 - or by subcontractors employed by the above
 - By Commissions / Agencies
 - Typically used for transportation infrastructure
 - Commissioners / board members appointed by government

Infrastructure

Government Owned Infrastructure Operators

- Government owned infrastructure may be operated in a number of ways:
 - By Authorities
 - Typically used for transportation infrastructure serving commercial carriers
 - Board members appointed from government and corporate stakeholders
 - Run as a not for profit
 - Expected to recover all costs
 - In some cases expected to lease land and self-fund capital improvements

Infrastructure

Government Owned Infrastructure Operators

- Government owned infrastructure may be operated in a number of ways:
 - By Crown Corporations
 - Typically used for utility infrastructure or transportation
 - Board members appointed by government
 - Run as a for profit or not for profit
 - Expected to recover all costs or subsidized
 - In some cases expected to lease land and self-fund capital improvements

Infrastructure

Privately Owned Infrastructure

- Infrastructure that serves the public good may be privately owned such as:
 - Gas / oil pipelines and distribution systems
 - Telecommunications systems
 - Toll Highways
 - Railways
 - Electrical utilities
- These infrastructure owners are typically for profit corporations

Infrastructure

Privately Owned Infrastructure

- Owners of privately held infrastructure may receive government benefits
 - Use of public or private lands (right of ways)
 - Law enforcement
 - Subsidies
- Private infrastructure owners are also subject to government regulation

Infrastructure

Public-Private Partnership

- Public-Private Partnership is a means of providing public services or public infrastructure
 - Risk is transferred or shared between partners

Infrastructure

Public-Private Partnership Models

- Operation & Maintenance Contract (O & M)
 - A private operator, operates a publicly-owned asset (e.g. water/wastewater treatment plant)
 - Ownership of the asset remains with the public entity
- Build-Finance
 - The private sector constructs an asset and finances the capital cost only during the construction period

Infrastructure

Public-Private Partnership Models

- Design-Build-Finance-Maintain (DBFM)
 - The private sector designs, builds and finances an asset
 - Provides hard facility management (hard fm) or maintenance services under a long-term agreement
- Design-Build-Finance-Maintain-Operate (DBFMO):
 - The private sector designs, builds, finances and provides hard fm or maintenance services under a long-term agreement
 - Operation of the asset is also included in projects such as bridges, roads and water treatment plants

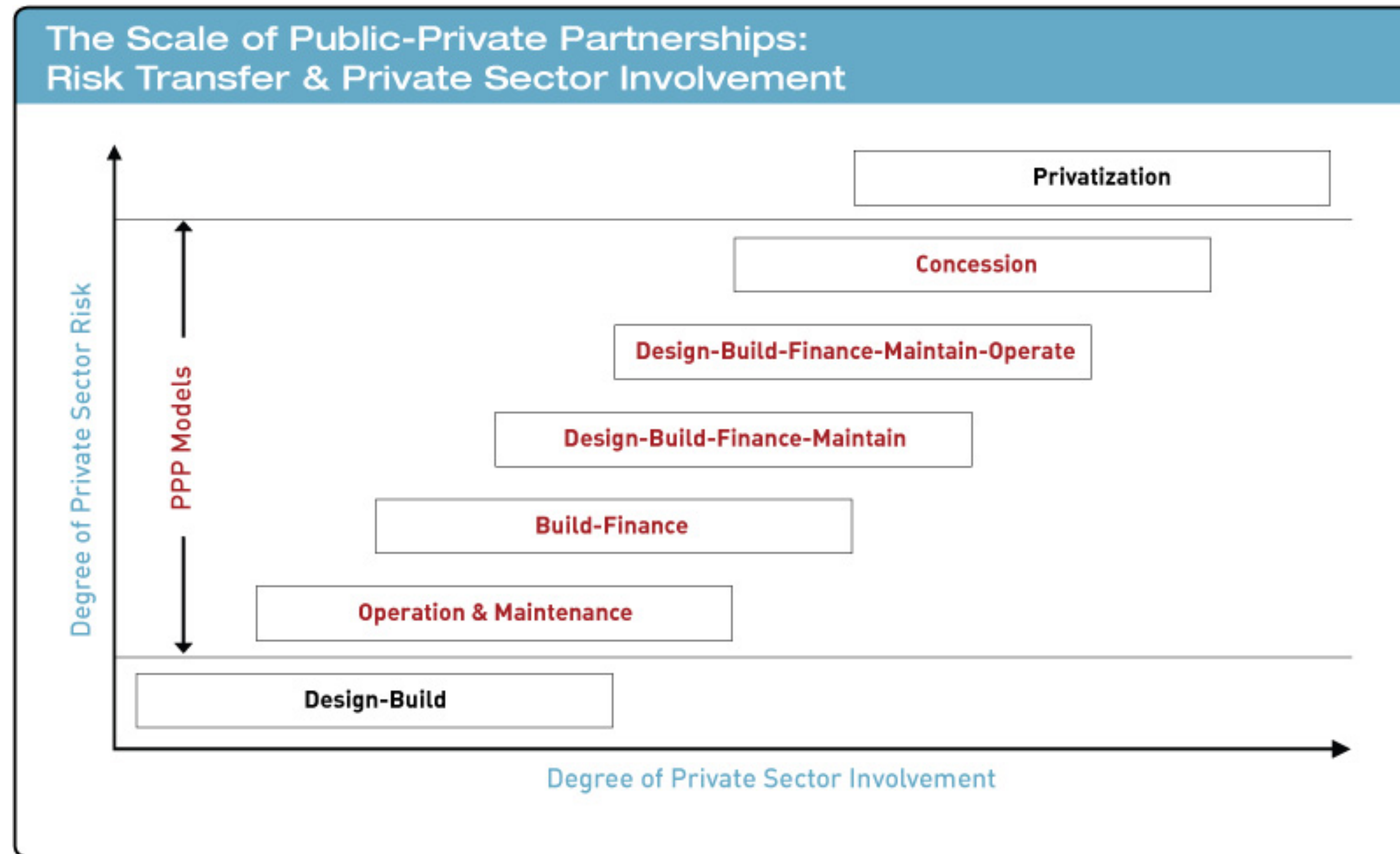
Infrastructure

Public-Private Partnership Models

- Concession
 - A private sector concessionaire undertakes investments and operates the facility for a fixed period of time
 - Ownership reverts back to the public sector.

Infrastructure

Public-Private Partnership Models



© The Canadian Council for Public-Private Partnerships

https://www.pppcouncil.ca/web/P3_Knowledge_Centre/About_P3s/Definitions_Models.aspx

Infrastructure

Public-Private Partnership Benefits

- Private sector actors have greater incentives to perform - the profit motive
- Overruns may be fully or partially absorbed by the private partner
- Penalty clauses or incentive clauses motivate performance

Infrastructure

Public-Private Partnership Criticisms

- Private sector partners pay lower wages / offer lower benefits
- Private sector borrowing costs are higher
- PPP direct public funds to corporate profits
- PPP projects exclude small local contractors in favour of large firms

Environmental Assessment

Purpose

- An environmental assessment is intended to guide government in approving public sector infrastructure projects (or permitting certain private sector projects such as mines)
- An environmental assessment
 - Identifies possible environmental effects
 - Proposes measures to mitigate adverse effects
 - Predicts whether there will be significant adverse environmental effects, even after the mitigation is implemented

Environmental Assessment

Scope

- In an Environmental Assessment (EA) the term *environment* is defined broadly to include:
 - The natural environment
 - The social environment
 - The cultural environment
 - The built environment
 - The economic environment

Environmental Assessment

Scope

- The Ontario Environmental Assessment Act defines the term *environment* as:
 - air, land or water;
 - plant and animal life, including human life
 - the social, economic and cultural conditions that influence the life of humans, or a community
 - any building, structure, machine or other device or thing made by humans
 - any solid, liquid, gas, odour, heat, sound, vibration or radiation resulting directly or indirectly from human activities
 - any part or combination of the foregoing and the interrelationships between any two or more of them

Environmental Assessment

Who is responsible for an EA?

- Various Environmental Assessment Acts (federal or provincial) lay out the types of project where an EA is required
- The proponent (the government department or organization advancing the project) is responsible for the EA
 - Many proponents use consultants to complete or support the EA process

Environmental Assessment

EA Outcomes

- The output of an EA is an Environmental Assessment Document
 - Sometimes called and Environmental Study Report (ESR) or Environment Impact Statement (EIS)
 - Typically includes a non technical summary for public understanding
- Typically the document is filed with the relevant ministry
- Documents are subject to
 - Public review and comment
 - Government (ministry) review

Environmental Assessment

The Role of Public Consultation

- Public consultation has been identified as a key component of the EA process
- EA Acts set out minimum consultation requirements
 - Many proponents chose to exceed these requirements

Environmental Assessment

The EA Process

- The full EA process in Ontario has 8 steps:
 1. Develop and submit a Terms of Reference
 2. Prepare an environmental assessment
 3. Submit an environmental assessment
 4. Public and government review

Environmental Assessment

The EA Process

- The full EA process in Ontario has 8 steps:
 5. Ministry of the Environment, Conservation and Parks review
 6. Public consultation on the Ministry Review
 7. Minister's decision
 8. Implement the project and monitor compliance

Environmental Assessment

The EA Process

- Terms of Reference include
 - The name and address of the proponent
 - How the environmental assessment will be prepared
 - Purpose of the study or undertaking
 - Description of and rationale for the undertaking and for alternatives
 - Description of the existing environment and potential effects of the undertaking

Environmental Assessment

The EA Process

- Terms of Reference include
 - Assessment and evaluation
 - Commitments and monitoring
 - Consultation plan for the environmental assessment
 - Flexibility to accommodate new circumstances
 - Other approvals required

Environmental Assessment

The EA Process

- The Environmental Assessment Document includes:
 - Record of consultation
 - Monitoring framework that will be carried out if the undertaking is approved
 - List of commitments
 - Actions to prevent, reduce and manage environmental effects

Environmental Assessment

The EA Process

- The Environmental Assessment Document includes:
 - Environmental effects that may be caused
 - Review and evaluation of alternatives considered
 - Results of the planning and decision-making process
 - Purpose of the project and a description of the undertaking

Environmental Assessment

Record of consultation

- Public Consultation
- Strategies
 - Notify people through the media reaching greatest number of people
 - Directly affected groups should be notified directly
 - Provide background info with the notice
 - Use non-technical language

Environmental Assessment

Actions to prevent, reduce and manage environmental effects

- Negative impacts on the environment identified during the EIA can be alleviated through mitigation measures
- Four general elimination/minimization strategies
 - Avoidance
 - Compensation
 - Mitigation (e.g. project design, scheduling)
 - Rectification
- Also consider managing positive impacts

Environmental Assessment

Environmental effects that may be caused

- Quantifying Impacts
- To meaning fully compare alternatives, a method must be used to quantify impacts
 - A rubric may be used to rate impact levels
 - An assessment matrix (weighted or unweighted) may be used to numerically analyze impacts

Environmental Assessment

Impact Rubric

Impact Significance	Impact Type		
	Physical	Biological	Socio-economic
Major	Factor affected over most of region for decades	Population of region affected for several generations	Whole population of region affected for several generations
Moderate	Factor affected over most of region for one or more decades	Portion of population affected for one or more generations	Community affected for one or more generations
Minor	Factor affected within most of region for less than 10 years	A specfic group of individuals within ecosystem affected for less than one generation	A specific group of individuals whin community are affected for less than one generation
Negligible	Factor affected within part of region for a short time	A specific group of individuals within region affected for a short time	A specific group of individuals within community affected for a short time

Environmental Assessment

Assessment Matrix - Unweighted

Environmental components	Project Activities							
	Plant Construction	Pesticide and Fertilizer Use	Raw materials transport	Water Intake	Solid Waste	Effluent Discharge	Emissions	Employment
Surface Water Quality		X			X	X		X
Surface Water Hydrology				X				
Air Quality			X				X	
Fisheries		X				X		
Terrestrial Wildlife Habitat	X							
Terrestrial Wildlife	X							
Land Use Pattern								
Highways/Railways			X					
Water Supply						X		
Agriculture		X				X		X
Housing								
Health					X	X	X	
Socioeconomic								X

Environmental Assessment

Assessment Matrix - Unweighted

		Impact Categories				
		Input			Output	
Company Processes		Input Materials	Energy	Waste	Liquid Emissions	Air Emissions
Production	Sand casting	1.00	2.00	4.00	2.00	1.00
	Sand core moulding	3.00	2.00	4.00	2.00	2.00
	Shell casting	2.00	3.00	4.00	2.00	2.00
	Shell core moulding	1.00	3.00	4.00	2.00	1.00
	Quality control	1.00	2.00	4.00	2.00	1.00
Maintenance		2.00	0.00	0.00	2.00	1.00
General services		0.00	1.00	1.00	2.00	1.00

Environmental Assessment

Assessment Matrix - Weighted

Environmental component	Weight	Fishery A		Fishery B		Fishery C	
		Impact	Score	Impact	Score	Impact	Score
Physical & chemical characteristics	20	-9	-180	-3	-60	-5	-100
Biological conditions	40	-2	-80	-5	-200	-4	-160
Ecological relationships	30	-5	-150	-4	-120	-6	-180
Cultural factors	10	3	30	3	30	4	40
Total	100		-380		-350		-400

Score = Impact x Weight

Environmental Assessment

Streamlined EA Process

- For many projects where the potential impacts can be inferred from similar projects a streamlined process may be allowed
- This may take the form of:
 - Class EAs
 - Project Specific Regulations
- The use of a streamlined EA does not negate the need for public consultation

Environmental Assessment

Benefits

- Reduced cost and time of project implementation
- Cost-saving modifications in project design
- Increased project acceptance
- Avoided impacts and violations of laws and regulations
- Improved project performance
- Avoided treatment/clean up costs

Environmental Assessment

Benefits

- Creating opportunities for meaningful public participation and engagement with Indigenous peoples
- Advancing and addressing any potential impacts to Indigenous rights
- Increasing protections for human health
- Reducing project costs and delays
- Reducing risks of environmental harm or disasters
- Lessening the probability that environmental effects transcend national, provincial or territorial boundaries
- Contributing to the responsible development of natural resources

Environmental Assessment

Criticism

- The scope of alternatives considered is not sufficiently broad
- The EA process is costly and slows projects
- The EA process can be used by opponents to derail projects